Temperature and particle motion

antocciphysics | TEACHER KEY | CPA

Learning target: Use a particle model to distinguish average motion from total energy.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 3; 3 course days later.*
Electromagnetic radiation can travel in a vacuum; conduction/convection require matter.

02. Explain a worked example (worked)
Warmer sample has greater average motion, not necessarily every particle faster than every colder one.

03. Look and explain (model)
Hotter sample has larger average kinetic energy; not every particle has the same speed.

04. Build the solution (practice)
Factor 2.

05. Use the idea (reason)
Must use kelvin: 293 to 313 K is only a factor 313/293≈1.068.

06. Choose a method (mixed) *Review: Unit 4, Day 16; 12 course days later.*
Same work; the slower student has 1/4 of the faster student’s average power.

07. Exit check (exit)
No; same average per particle, roughly twice total translational energy for the same ideal gas conditions.

08. STRETCH: Apply it another way (stretch)
No; temperature relates to average kinetic energy, not volume alone. Mass/species also affects typical speeds.

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 3; 3 course days later.*
Net thermal radiation transfers energy to the cooler radiative environment. Absolute zero is not required.

02. Explain a worked example (worked)
Warmer sample has greater average motion, not necessarily every particle faster than every colder one.

03. Represent and interpret (model)
Hotter sample has larger average kinetic energy; not every particle has the same speed.

04. Solve and justify (practice)
6.21×10^{-21} J; doubles to 1.242×10^{-20} J.

05. Analyze the reasoning (reason)
Must use kelvin: 293 to 313 K is only a factor 313/293≈1.068.

06. Choose a method (mixed) *Review: Unit 4, Day 16; 12 course days later.*
P=Fv=240 W; W=Pt=1920 J.

07. Exit check (exit)
√2, since kinetic energy is proportional to speed squared.

08. STRETCH: Apply it another way (stretch)
Speed doubles; the relationship uses absolute temperature. Celsius zero is not zero thermal motion.